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First Direct Structural Evidence for Age-Dependent Polyglutamylation Asymmetry in Murine Hematopoietic Stem Cells

A Phase 0 Protocol and Predicted Outcomes

Author: Jaba Tkemaladze
Affiliation: Georgia Longevity Alliance, Reg. №404506520
Correspondence: jaba@longevity.ge | ORCID: 0000-0002-3826-7982
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Abstract

Asymmetric partitioning of cellular determinants, including post-translational modifications (PTMs), is a proposed mechanism for the aging of somatic stem cells. The Centriolar Damage Accumulation Theory of Aging (CDATA) posits that the asymmetric segregation of a polyglutamylated mother centriole protects the stem cell daughter, with age-related decline in this asymmetry driving functional attrition. While indirect evidence supports this in other systems, no study has directly visualized polyglutamylation asymmetry in primary hematopoietic stem cells (HSCs). Here, we present a Phase 0 protocol to obtain the first structural evidence for age-dependent polyglutamylation asymmetry in lineage-negative, Sca-1+, c-Kit+ (LSK) cells from young and aged mice. We describe a three-level correlative approach: 1) quantifying mother/daughter centriole polyglutamylation via GT335/Ninein co-immunostaining, 2) assessing primary cilium frequency (ARL13B), and 3) measuring ex vivo division kinetics and co-culture "relapse" dynamics. We provide five quantitative, falsifiable predictions: aged LSK will show a higher asymmetry index, lower ciliation frequency, and reduced division rates, with transient co-culture rescue of aged cell proliferation followed by relapse coincident with re-accumulation of polyglutamylation. This pre-registered design establishes a falsifiable baseline for CDATA in hematopoiesis and ensures the foundational structural data are disseminated independently of subsequent functional validation.

Keywords: hematopoietic stem cells, centriole, polyglutamylation, asymmetry, aging, GT335, CDATA, Phase 0 protocol

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Introduciton

The decline in regenerative capacity of the hematopoietic system with age is intrinsically linked to functional exhaustion of hematopoietic stem cells (HSCs) (Wilson et al., 2008). This exhaustion manifests as myeloid-biased differentiation, reduced engraftment potential, and increased cellular stress (Lavasani et al., 2012; Leins et al., 2022). A central, unresolved question is the molecular mechanism driving this attrition. The asymmetric division hypothesis suggests that somatic stem cells can unequally partition damaged components to generate one "rejuvenated" self-renewing daughter and one committed or aged daughter (Yamashita et al., 2007; Rando & Royall, 2025). The centrosome, as a microtubule-organizing center, is a prime candidate for orchestrating such asymmetry.

The Centriolar Damage Accumulation Theory of Aging (CDATA) provides a specific, testable molecular framework for this process (Tkemaladze, 2023, 2026). It proposes that the mother centriole, distinguished by distal appendage proteins like Ninein, accumulates tubulin polyglutamylation—a PTM linked to microtubule stability and ciliary function. During asymmetric HSC division, the hypothesis states that the highly polyglutamylated mother centriole is selectively retained by the self-renewing daughter stem cell, while the daughter centriole, with lower polyglutamylation, is allocated to the differentiating cell. This asymmetry is proposed to protect the stem cell lineage from the cumulative burden of centriole-associated damage. Age-related erosion of this asymmetry would lead to a more symmetric distribution of "aged" centrioles, propagating damage and compromising stemness.

Compelling indirect evidence supports elements of CDATA. In *Drosophila* germline stem cells, centrosome asymmetry regulates fate (Yamashita et al., 2007; Cheng & Yamashita, 2008). In murine neural progenitors and activated CD8+ T-cells, centriolar age influences cell fate and division patterns (Royall et al., 2023; Barandun et al., 2025). Crucially, the core structural prediction—asymmetric enrichment of polyglutamylation on the mother centriole—and its age-dependent dysregulation have **never been directly visualized or quantified in primary HSCs**. All existing inferences rely on surrogate markers or non-hematopoietic systems. The Lin−Sca-1+c-Kit+ (LSK) compartment, containing long-term HSCs, is the definitive physiological context for this question. Cell lines (e.g., KG-1a, 32D) fail to recapitulate primary HSC centrosome biology and division patterns, creating a critical gap in the field.

This study presents a Phase 0 protocol designed to acquire the first direct structural evidence for polyglutamylation asymmetry in primary murine HSCs/early progenitors (LSK cells) and to test key functional corollaries of CDATA. By publishing this hypothesis-driven design with explicit, quantitative predictions prior to data collection, we aim to: 1) establish an unbiased, falsifiable framework for testing CDATA in hematopoiesis, 2) ensure that foundational structural data are disseminated regardless of future funding outcomes, and 3) provide a predictive model that can be directly validated or refuted by the community.

Methods (Phase 0 Protocol)

Animals and Cell Isolation

Young (2-3 months) and aged (20-24 months) male C57BL/6J mice (n=5 per group per experiment) will be used. Bone marrow will be harvested from femurs and tibias. Lineage-negative cells will be enriched using a biotinylated lineage antibody cocktail (Ter119, CD3e, CD11b, Gr-1, B220) and anti-biotin microbeads. LSK cells will be isolated via fluorescence-activated cell sorting (FACS) for Lin−, Sca-1+, c-Kit+ population. GFP+ transgenic mice (C57BL/6-Tg(UBC-GFP)30Scha/J) will be used as a source of young GFP+ LSK cells for co-culture experiments.

Level 1 – Structural Analysis of Centriolar Polyglutamylation Asymmetry (Estimated Cost: ~$6,000)

Sorted LSK cells will be cytospun onto slides, fixed in cold methanol, and permeabilized. Immunofluorescence will be performed using: i) mouse monoclonal GT335 antibody (1:500, AdipoGen, recognizes polyglutamylated tubulin), ii) rabbit polyclonal anti-Ninein antibody (1:250, Abcam, mother centriole marker), and iii) DAPI. Secondary antibodies (Alexa Fluor 488 anti-mouse, Alexa Fluor 594 anti-rabbit) will be used.

Specificity Controls for GT335:

- *Positive control:* Mouse brain sections (known high polyglutamylation levels) will be stained in parallel
- *Negative control:* GT335 pre-incubated with excess polyglutamylated peptide (competition assay)
- *Secondary-only control:* Omission of primary antibody to assess background fluorescence

High-resolution z-stack images will be acquired on a confocal microscope (63x/1.4 NA oil objective). **All image analysis will be performed blinded to age group** (slides coded by an independent technician). For each cell with clearly resolved centriole pairs (n≥50 per age group), the Mean Fluorescence Intensity (MFI) of the GT335 signal will be quantified on the Ninein-positive (mother) and Ninein-negative (daughter) centriole using ImageJ.

Cell Exclusion Criteria:

- Cells with ambiguous or overlapping centriole signals
- Cells with damaged or fragmented nuclei (DAPI staining)
- Cells with clearly disrupted cytoplasmic membrane
- Cells with fewer than two resolvable centrioles
- More than 10% of cells per sample will be excluded; if >20% are excluded, the sample will be re-processed

The **Asymmetry Index (AI)** will be calculated as: AI = MFI(Ninein+) / MFI(Ninein−). An AI > 1 indicates asymmetric polyglutamylation enrichment on the mother centriole.

Limitation Note: GT335 recognizes polyglutamylated tubulin but does not distinguish between different chain lengths (short vs. long glutamate chains). This distinction may be biologically relevant, as different TTLL enzymes add different chain lengths with potentially distinct functions.

Level 2 – Analysis of Primary Cilium Frequency (Estimated Cost: ~$3,000)

Given the mother centriole's role as the basal body for the primary cilium—a signaling organelle—CDATA predicts ciliary defects with aging. Cytospun LSK cells will be stained for ARL13B (mouse monoclonal, 1:250, NeuroMab, ciliary membrane marker) and Ninein. The frequency of LSK cells possessing an ARL13B+ primary cilium (co-localized with a Ninein+ centriole) will be determined from ≥100 cells per age group.

Supporting Evidence: Previous work has demonstrated that senescent mesenchymal stem cells (MSCs) exhibit reduced primary cilia frequency and decreased ARL13B intensity, accompanied by impaired Hedgehog signaling (Hindawi/Wiley, 2021). This provides a precedent for P2 in the stem cell aging field.

Level 3 – Functional Division Kinetics and Co-culture "Relapse" Assay (Estimated Cost: ~$8,000)

Division Rate: Sorted young and aged LSK cells will be cultured in StemSpan SFEM II with cytokines (SCF, TPO, FLT3-L). Ki67 staining and EdU incorporation (Click-iT EdU kit) will assess the percentage of cells in cycle at 48h and 72h.

Relapse Co-culture: Aged (GFP−) and young (GFP+) LSK cells will be co-cultured at a 1:1 ratio in the above conditions. This design tests if exposure to a "young" secretome/microenvironment can transiently reset aged HSC division kinetics. Cells will be sampled at T=0 (pre-culture), and after approximately 20, 60, and 100 cumulative population doublings (tracked via cell counts). At each timepoint, GFP− (aged) cells will be analyzed for: a) Ki67 positivity (flow cytometry), and b) GT335 MFI (via immunofluorescence after sorting).

Statistical Power: With n=5 per group, assuming Cohen's d ≈ 1.5-2.0 (based on pilot data from other systems), the study has >80% power to detect significant differences at $\alpha=0.05$.

Quantitative Predictions & Falsification Criteria

We state the following *a priori* predictions derived from CDATA theory (Tkemaladze, 2026) and the Cell-DT computational model (Cell-DT, 2024). The experiment will be considered to falsify CDATA's applicability to HSCs if the primary prediction (P1) is not supported.

P1: Asymmetry Index (AI) is higher in aged LSK cells.

- *Prediction:* AI_young < AI_aged

- *Rationale:* This counterintuitive prediction arises because CDATA posits that aged HSCs have accumulated more total centriolar polyglutamylation, and while asymmetry is eroded, the *absolute difference* in PTM load between mother and daughter centrioles may be larger.
- *Falsification Criterion:* Non-significant difference ($p \geq 0.05$, two-tailed t-test) or AI_young ≥ AI_aged

P2: Primary cilium frequency is reduced in aged LSK cells.

- *Prediction:* % ARL13B+ cells_aged < % ARL13B+ cells_young
- *Rationale:* Supporting evidence from senescent MSCs shows reduced ciliation frequency (Hindawi/Wiley, 2021)
- *Falsification Criterion:* No significant difference in ciliation frequency between age groups

P3: Aged LSK cells exhibit reduced ex vivo division rates.

- *Prediction:* % Ki67+ or EdU+ cells at 48/72h is lower in aged LSK cultures
- *Falsification Criterion:* No significant difference in proliferation indices

P4: Co-culture induces transient rescue of aged cell proliferation, followed by relapse.

- *Prediction:* In co-culture, the Ki67+ percentage in GFP− (aged) cells will increase significantly by ~T+20 doublings (rescue), begin declining by T+60, and return to near baseline (T=0) levels by T+100 (relapse)
- *Falsification Criterion:* Ki67 in GFP− cells remains stably rescued at T+100 or shows no initial rescue

P5: Polyglutamylation continues accumulating in aged cells during co-culture.

- *Prediction:* GT335 MFI (total cellular or centriolar) in sorted GFP− cells at T+100 > GT335 MFI at T=0
- *Falsification Criterion:* GT335 signal does not increase over the co-culture period

Expected Impact

This Phase 0 protocol is poised to generate the first HSC-specific structural dataset on centriolar polyglutamylation asymmetry, filling the critical evidence gap between CDATA theory and hematopoietic aging. The publication of this design and its outcomes—whether confirmatory or falsifying—has intrinsic scientific value. A positive result would provide direct visual proof of a key CDATA tenet in the relevant primary cell type, immediately elevating the hypothesis and guiding targeted functional studies (e.g., manipulating TTLL enzymes). A negative result would definitively falsify CDATA's predictions for HSCs, preventing future resource misallocation and redirecting the field toward alternative mechanisms of asymmetry.

Furthermore, this work operationalizes the "Registered Report" model for hypothesis-driven stem cell biology. By divorcing the narrative from post-hoc interpretation, it mitigates bias and enhances reproducibility. The co-culture "relapse" experiment, in particular, tests a novel, dynamic prediction of stem cell aging that moves beyond static young-vs-old comparisons.

Discussion

This protocol is deliberately focused on primary LSK cells, as cell lines (e.g., KG-1a, 32D) are inadequate models for HSC centrosome biology. These lines often exhibit aberrant centriole numbers, disrupted cell cycle regulation, and lack the authentic asymmetric division patterns of quiescent HSCs. Data from such lines would be uninterpretable in the context of a theory about *stem cell* aging *in vivo*.

Key Methodological Safeguards:

- Blinded image analysis to eliminate observer bias
- Specificity controls for GT335 (positive, negative, secondary-only)
- Pre-defined cell exclusion criteria (≥10% exclusion threshold with re-processing rule)
- Pre-registered falsification criteria for all five predictions

Key Limitations Acknowledged:

1. **GT335 specificity:** This antibody recognizes polyglutamylated tubulin but does not distinguish between different chain lengths (short vs. long glutamate chains). Different TTLL enzymes add different chain lengths with potentially distinct functional consequences. Future studies may require chain-specific antibodies or mass spectrometry approaches.
2. **In vitro vs. in vivo:** Division kinetics and co-culture assays are performed ex vivo, which may not fully recapitulate the bone marrow niche environment.
3. **Sample size:** n=5 per group is sufficient for structural analysis but may be underpowered for detecting small effect sizes in functional assays (P3-P5).
4. **AI validation:** The Asymmetry Index should ideally be validated using an independent method (e.g., mass spectrometry on isolated centrioles) in future studies.

The decision to publish a Phase 0 protocol with predictions is justified by the foundational nature of the proposed structural data. Whether the subsequent, more costly Phase 1 (e.g., in vivo transplantation with centriole tracking) is funded, the community will have access to the core evidence for or against polyglutamylation asymmetry in HSCs. This aligns with open science principles and accelerates the field's progress.

The predictions are quantitatively precise and falsifiable. Notably, P1 (higher AI in aged cells) is a non-obvious, counter-intuitive prediction that provides a strong test of the theory's mechanistic nuance. The "relapse" co-culture experiment (P4, P5) tests the persistence of cell-intrinsic aging

signatures against microenvironmental modulation, a key frontier in aging research (Rando & Royall, 2025).

Finally, these results have the potential to formally update the Bayesian Hypothesis Confidence Assessment (BHCA) for CDATA. The current BHCA, based on evidence from non-HSC systems, stands at approximately 17 out of 27 possible confidence points (Tkemaladze, 2026). Successful confirmation of all five predictions here, which address the most direct structural and initial functional correlates in the primary target cell, would project an increase to **22/27 points**. This would represent a significant consolidation of the theory's evidentiary basis and strongly justify large-scale investment in functional perturbation studies in vivo.

In conclusion, this Phase 0 protocol defines a critical, missing experiment in the field of hematopoietic stem cell aging. Its execution will provide definitive structural evidence for or against age-dependent centriolar PTM asymmetry, establishing a rigorous, falsifiable benchmark for all future work on CDATA and asymmetric inheritance in HSCs.

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